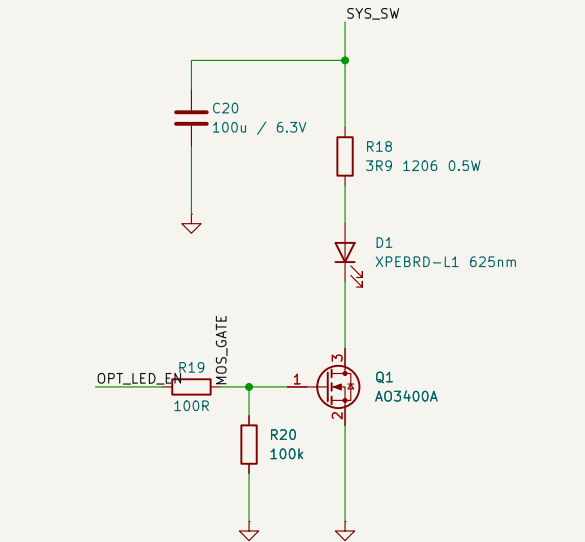


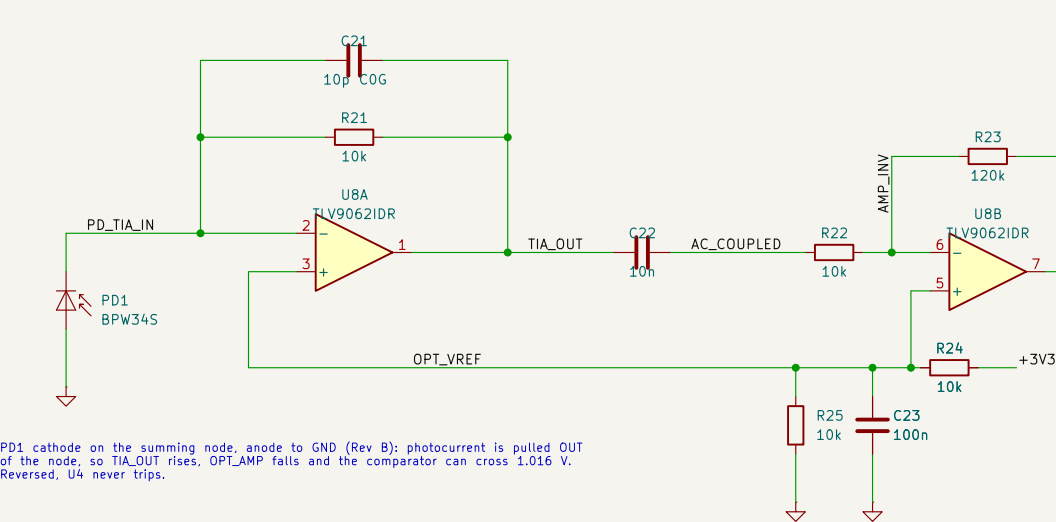
PULSED EMITTER – 20 kHz / 5  $\mu$ s / ~500 mA

Signal flows left to right: emitter, detector, gain, comparator, cable.



Emitter loop is local: C20 supplies the 5  $\mu$ s / 500 mA pulse so the cable carries only the ~50 mA average. R19 damps the gate edge; R20 holds Q1 off while the main board is unpowered or the cable is unplugged.

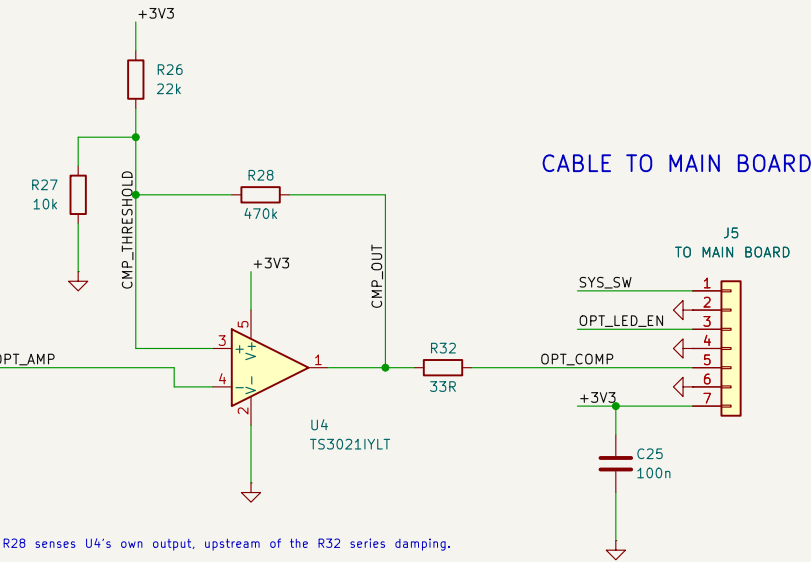
DETECTOR > TRANSIMPEDANCE > x12 AC GAIN



PD1 cathode on the summing node, anode to GND (Rev.B): photocurrent is pulled OUT of the node, so TIA\_OUT rises, OPT\_AMP falls and the comparator can cross 1.016 V. Reversed, U4 never trips.

1.65 V REFERENCE

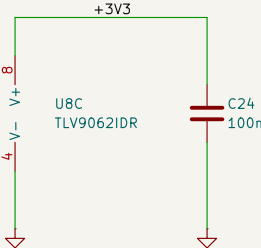
THRESHOLD COMPARATOR + HYSTERESIS



R28 senses U4's own output, upstream of the R32 series damping.

Ground sits between OPT\_LED\_EN and OPT\_COMP. Emitter-pulse crosstalk onto OPT\_COMP lands inside the synchronous gating window and would be accepted as a genuine hit, so it must be stopped here, not filtered downstream. Use a latching, gold-plated JST series with secondary retention: tin friction contacts fret under sustained vibration and fail intermittently rather than cleanly.

SUPPLY + ERC FLAGS



BalancerREF\_OptHead – optical tach head, at 90 degrees to the main PCB on the aluminium L-bracket. One pivot bolt aims the emitter; add a second fastener (arc slot or serrated lock nut) or it drifts under vibration and loses RPM lock. The baffle between D1 and PD1 and the red acrylic window belong to this board. Designators match BalancerREF Rev C so the existing pin audits stay valid.

Cable carries DC rails and digital edges only  
Split from BalancerREF Rev C – human review required

Sheet: /  
File: BalancerREF\_OptHead.kicad\_sch

Title: BalancerREF\_OptHead – pivoting optical tach head

Size: A3 Date: 2026-09-11 Rev: B  
KiCad E.D.A. 10.0.3 Id: 1/1